Five-Year Career Development Plan (2026–2031)

Mathematics and Software Engineering Pathway

I am currently a **200-level Pure Mathematics student** at Lagos State University, expected to graduate in **2028**. My long-term objective is to integrate rigorous mathematical training with professional software engineering experience, eventually transitioning into Machine Learning, Data Science, or advanced computational research.

This five-year plan outlines my structured progression from undergraduate training to industry experience and, ultimately, advanced specialization.

Phase I — Foundation Stage (Final Undergraduate Years: 2026–2028)

Primary Objective

To graduate not only with a Mathematics degree but as an **industry-ready Software Engineer** possessing practical development experience and a strong technical portfolio.

Year 1: Skill Consolidation (Current Level → 300 Level)

Software Engineering Development

Focus on becoming a strong backend and full-stack engineer by mastering:

- NestJS architecture and backend design principles
- Authentication and authorization systems
- PostgreSQL database management
- Redis caching strategies
- RESTful API design and system architecture fundamentals
- ReactJs (NextJs) TanStack ecosystem for advanced frontend data management.

Key Deliverables

- Development of a production-level SaaS application
- Design and deployment of a scalable backend system

Mathematical Development

Strengthen theoretical foundations essential for future computational work:

- Linear Algebra
- Probability Theory
- Real Analysis

- Graph Theory
- Optimization fundamentals

These areas will form the mathematical basis for future Machine Learning and Data Science specialization.

Professional Positioning

- Build and maintain a strong GitHub portfolio
- Publish technical articles or engineering write-ups
- Develop professional presence on LinkedIn
- Apply for internships, startup collaborations, and open-source contributions

Target Outcome:
Secure first paid internship or freelance software role before entering final year.

Year 2: Industry Readiness (Final Year — 2027/2028)

Engineering Focus

Transition from student developer to employable engineer through:

- Backend scalability techniques
- Docker and containerization basics
- Cloud deployment workflows
- CI/CD pipeline implementation
- Preparation for system design interviews

Target Role:
Junior Full-Stack Software Engineer (part-time or entry level).

Graduation Goal

Complete undergraduate studies with:

- Professional work experience
- A strong technical portfolio
- Active income stream
- Established industry network

Phase II — Industry Experience Stage (Professional Engineering Period: 2028–2031)

Before pursuing postgraduate studies, I plan to gain substantial industry exposure as a software engineer, allowing me to develop practical expertise, financial stability, and clearer research direction.

Year 1 Post-Graduation (2028–2029)

Professional Identity: Full-Stack Software Engineer

Focus Areas

- Working within production engineering teams
- Managing large codebases
- Collaborative software development practices
- Understanding real-world software architecture

Skills Development

- Testing culture and software reliability
- Performance optimization
- Microservices architecture
- Production debugging and maintenance

Outcome:
Strong professional foundation and real industry experience.

Year 2 Post-Graduation (2029–2030)

Professional Identity: Backend / Systems Engineer

Expansion Areas

- Python programming for computation and data processing
- Data structures and algorithmic problem solving
- Distributed systems concepts
- Message queues and event-driven architecture
- Data pipelines and backend data engineering

Initial Exposure

- Data Analytics workflows
- Introductory Machine Learning concepts

Year 3 Post-Graduation (2030–2031)

Transition Stage — Advanced Computing

At this stage, I will combine:

- Mathematical training
- Professional engineering experience
- Financial independence

New Focus

- Machine Learning and Artificial Intelligence
- Data Science applications
- Mathematical modeling and computational research

Key Activities

- Development of Machine Learning projects
- Research-oriented technical work
- Technical publications and portfolio expansion

Next Step
Application to MSc programs or transition into AI/ML or Research Engineering roles.

Long-Term Vision

My overarching goal is to evolve into a professional capable of bridging:

Pure Mathematics → Software Engineering → Artificial Intelligence → Research Innovation

By intentionally gaining industry experience before postgraduate study, I aim to approach advanced education with practical insight, technical maturity, and clear research direction.